Activity-Based Assessment: Generative AI in Gaming

Maximum Marks: 10

1. Objective

Design and develop a playable game prototype that integrates a Generative AI tool to generate at least one game element, such as dialogue, questions, stories, clues, characters, or game scenarios.

The objective is to demonstrate how Generative AI can make games more interactive, dynamic, and engaging.

2. Problem Statement

Develop a working prototype of a game that uses Generative AI to generate content based on the player's input, choices, or progress in the game.

You may choose any one of the following concepts:

- **AI Story Adventure:** Generate a story that changes according to the player's decisions.
- **Non-Player Character (NPC):** Create an AI-powered character that interacts with the player through dynamically generated dialogue.
- **Quiz Game:** Generate questions, hints, and feedback dynamically based on the player's answers.
- **Simulation Game:** Generate situations or scenarios that require the player to make decisions.
- **Mystery or Detective Game:** Generate clues, suspects, and story developments for the player to investigate.

You may propose a different game concept, provided it meets the assessment requirements.

3. Technical Requirements

Your prototype must satisfy the following requirements:

1. **Playable game:** The user must be able to interact with the game through choices, answers, commands, or other controls.
2. **Generative AI integration:** Use a Generative AI tool or model to generate at least one game element at runtime.

3. **Player interaction:** The game must respond to the player's actions or inputs.
4. **Working demonstration:** Demonstrate the game from beginning to at least one meaningful outcome.
5. **Error handling:** Include a basic mechanism to handle invalid input or AI/API errors, where applicable.

You may use Python, JavaScript, HTML/CSS, a game engine, or another suitable development platform. You may integrate an AI API or use an appropriate AI-enabled tool.

Important: A static presentation, a video of a game, or a collection of predefined prompts and responses alone will not qualify as a working prototype. The Generative AI component must be integrated into the playable experience.

4. Submission Requirements

Submit the following:

- Working prototype or project files.
- A short README explaining the game concept, tools used, and how to run the game.
- A brief explanation of where and how Generative AI is used.
- A short demonstration of the game during evaluation.

Be prepared to explain your implementation and demonstrate that the game works.

5. Evaluation Rubric — 10 Marks

Assessment criterion	**Marks**
Game concept, creativity, and relevance	2
Integration of Generative AI	3
Playability and player interaction	2
Correctness and working demonstration	2
Explanation of implementation	1
Total	**10**

6. Guidelines

- Work individually or in groups, as instructed by the faculty.

- Keep the scope small and achievable. A simple, functioning game is better than an ambitious but incomplete project.
- You may use Generative AI tools to assist with coding, but you must understand and explain your implementation.
- Do not expose API keys or other credentials in submitted source code.
- Clearly identify the content generated by the AI.
- Test the prototype before the evaluation.

Expected outcome: A playable prototype in which Generative AI contributes to the gaming experience rather than being used only to write the code.